



Modelling Financial Markets

Chapter 5: Conditional Volatility — GARCH Models



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Acronyms used in this material (1/2)

- ▣ ACF: AutoCorrelation Function
- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ AR: AutoRegressive (model)
- ▣ ARCH: AutoRegressive Conditional Heteroskedasticity
- ▣ ARIMA: AutoRegressive Integrated Moving Average (model)
- ▣ ARMA: AutoRegressive Moving Average (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-TR: BET Total Return (index)
- ▣ BIC: Bayesian Information Criterion
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ DCC: Dynamic Conditional Correlation
- ▣ DM: Diebold–Mariano (test)
- ▣ EGARCH: Exponential GARCH
- ▣ EWMA: Exponentially Weighted Moving Average
- ▣ FIGARCH: Fractionally Integrated GARCH
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GARCH-M: GARCH-in-Mean
- ▣ GARCH-MIDAS: GARCH with Mixed Data Sampling (long-run component)
- ▣ GARCH-N: GARCH with Normal innovations
- ▣ GAS: Generalised Autoregressive Score (model)
- ▣ GED: Generalised Error Distribution
- ▣ GJR: Glosten–Jagannathan–Runkle (asymmetric GARCH)
- ▣ GJR-GARCH: Glosten–Jagannathan–Runkle GARCH
- ▣ GW: Giacomini–White (test)

Acronyms used in this material (2/2)

- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HAR: Heterogeneous AutoRegressive (model)
- ▣ IGARCH: Integrated GARCH
- ▣ LM: Lagrange Multiplier
- ▣ LR: Likelihood Ratio (test)
- ▣ MCS: Model Confidence Set
- ▣ MLE: Maximum Likelihood Estimation
- ▣ MSE: Mean Squared Error
- ▣ MZ: Mincer–Zarnowitz (regression)
- ▣ PIT: Probability Integral Transform
- ▣ QLIKE: Quasi-LIKelihood (loss function)
- ▣ QMLE: Quasi-Maximum Likelihood Estimator
- ▣ QQ: Quantile–Quantile (plot)
- ▣ QSD: Quasi Score-Driven (model)
- ▣ VIX: CBOE Volatility Index
- ▣ XAU: gold, ISO 4217 code (one troy ounce)

Chapter Route and Outcomes

Three questions (2 × 80 minutes)

- How does the variance evolve? ARCH, GARCH, persistence, forecasts (Session 1)
- How are the models estimated and checked? Likelihood, QMLE inference, boundary tests, asymmetry, diagnostics (Sessions 1–2)
- Do the forecasts improve? Out-of-sample losses, tests for many models, limits and extensions, a case study (Session 2)
- Technical detail sits in the Appendix at the end

After this chapter you can

- estimate GARCH, GJR and EGARCH with heavy-tailed innovations and read every parameter
- attach valid uncertainty to persistence, half-life and tests on the boundary of the parameter space
- diagnose standardised residuals and compare variance forecasts with proxy-robust losses

Volatility Comes in Bursts

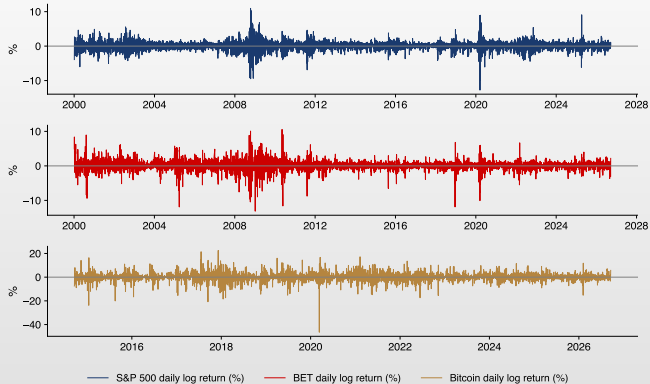


Amsterdam Stock Exchange, 21 October 1987, after the crash of 19 October

Photo: Bart Molendijk / Anefo (1987); CC0; Wikimedia Commons

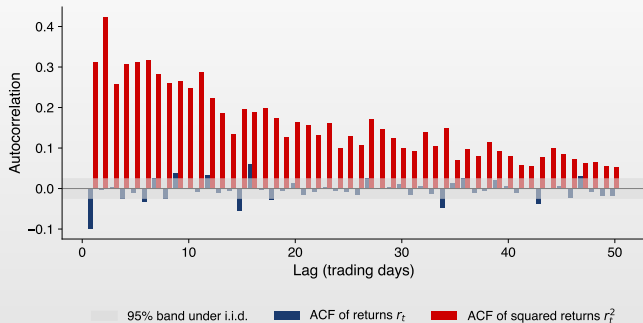
- Crashes are not confined to isolated days: large moves are followed by large moves
 - ▶ “Large changes tend to be followed by large changes, of either sign” [Mandelbrot \(1963\)](#)
- This **volatility clustering** is stylised fact 6 of Chapter 1
- Volatility enters every risk number: Value at Risk (Chapter 7), portfolio weights (Chapter 4), option prices
- Question: can we turn the clustering into a **forecast** of tomorrow’s risk, and conduct valid inference on it?

Volatility Clusters in Every Market



- Daily log returns $r_t = 100 (\ln P_t - \ln P_{t-1})$ in %, closes up to 18 September 2026; S&P 500 and BET from 2000, Bitcoin from 17 September 2014
- Equity indices: weekdays; days whose close equals the previous close (holidays filled with the last price) are dropped: 3 for the S&P 500, 0 for the BET; Bitcoin: 7 days a week
- Quiet years alternate with turbulent months in all three: 2008, March 2020, April 2025

The Sign Is Unpredictable, the Size Is Not



- S&P 500, lag 1: -0.099 for r_t , 0.313 for r_t^2 ; all 50 autocorrelations of r_t^2 up to lag 50 lie outside ± 0.024
- Heavy tails too: excess kurtosis 10.7 (S&P 500), 11.1 (BET), 12.0 (Bitcoin); 0 for the Normal distribution
- Object to model: $\sigma_t^2 = \text{Var}(r_t \mid \mathcal{F}_{t-1})$, with $\mathcal{F}_{t-1}$ the information at the close of day $t - 1$

Testing for ARCH Effects

- H_0 : no clustering, $\text{Var}(r_t | \mathcal{F}_{t-1})$ constant
- **McLeod–Li** [McLeod & Li \(1983\)](#): Ljung–Box $Q(10)$ on r_t^2 ; reject if $Q(10) > 18.3$ (χ_{10}^2 , 5%)
- **Engle's LM test** [Engle \(1982\)](#): regress $\hat{\varepsilon}_t^2$ on a constant and $\hat{\varepsilon}_{t-1}^2, \dots, \hat{\varepsilon}_{t-5}^2$, $\hat{\varepsilon}_t = r_t - \bar{r}$
 - ▶ $LM = nR^2$; H_0 : all five slopes are 0; reject if $LM > 11.1$ (χ_5^2 , 5%)
- Every statistic is far above its critical value (all $p < 0.001$): clustering in equities, crypto, a managed exchange rate and gold; weakest for Bitcoin and gold
- One-sided score tests and the full table: Appendix

Series	$Q(10)$ on r^2	LM(5)	$\hat{\rho}_1(r^2)$
S&P 500	6133	1684	0.313
BET	2440	982	0.330
BET-TR	714	371	0.171
Bitcoin	213	115	0.137
EUR/RON	2420	871	0.342
Gold	744	329	0.110

The ARCH Model of Engle (1982)

- ▣ Setting: $r_t = \mu + \varepsilon_t$, $\varepsilon_t = \sigma_t z_t$, innovations z_t i.i.d. with mean 0 and variance 1
 - ▶ Conditional variance $\sigma_t^2 = \text{Var}(r_t | \mathcal{F}_{t-1})$; unconditional variance $\sigma^2 = \mathbb{E}\sigma_t^2$
- ▣ **ARCH(q)**: $\sigma_t^2 = \omega + \alpha_1 \varepsilon_{t-1}^2 + \cdots + \alpha_q \varepsilon_{t-q}^2$ Engle (1982)
 - ▶ $\omega > 0$, $\alpha_i \geq 0$: the variance is always positive
- ▣ A large shock yesterday (ε_{t-1}^2 large) raises today's variance: clustering by construction
- ▣ ARCH(1): unconditional variance $\sigma^2 = \omega/(1 - \alpha_1)$, finite if $\alpha_1 < 1$; its fourth moment is the case $\beta = 0$ of the GARCH result below
- ▣ Engle received the 2003 Nobel Prize for ARCH (history: Appendix)

From ARCH to GARCH (Bollerslev, 1986)

- Problem with ARCH: clustering in real data lasts for months, so q must be large (many parameters)
- **GARCH(1,1)** Bollerslev (1986): $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$
 - ▶ $\omega > 0, \alpha \geq 0, \beta \geq 0$; α : reaction to news; β : memory of past variance
- Substituting backwards: $\sigma_t^2 = \frac{\omega}{1-\beta} + \alpha \sum_{j \geq 1} \beta^{j-1} \varepsilon_{t-j}^2$
 - ▶ An ARCH(∞) with geometrically declining weights, using only three parameters
- GARCH(p, q) adds more lags; in practice GARCH(1,1) is the workhorse Hansen & Lunde (2005)

GARCH(1,1) as an ARMA Model for Squared Shocks

- Define $\nu_t = \varepsilon_t^2 - \sigma_t^2$, the surprise in the squared shock ($\mathbb{E}[\nu_t | \mathcal{F}_{t-1}] = 0$)
- Then $\varepsilon_t^2 = \omega + (\alpha + \beta) \varepsilon_{t-1}^2 + \nu_t - \beta \nu_{t-1}$
 - ▶ An ARMA(1,1) in ε_t^2 with autoregressive coefficient $\alpha + \beta$
- Consequence: the autocorrelations of ε_t^2 decay geometrically, $\rho_k = \rho_1(\alpha + \beta)^{k-1}$
 - ▶ $\rho_1 = \alpha(1 - \beta^2 - \alpha\beta)/(1 - \beta^2 - 2\alpha\beta)$ if $\mathbb{E}\varepsilon_t^4 < \infty$ (Seminar 5, A6)
 - ▶ S&P 500, GARCH(1,1) with Normal innovations: $\rho_1 = 0.356$, $\rho_{10} = 0.298$; sample 0.313 and 0.248
- $\alpha + \beta$ is therefore called the **persistence** of volatility

Stationarity and the Unconditional Variance

- ▣ **Covariance stationarity:** $\alpha + \beta < 1$; then

- ▶ $\sigma^2 = \mathbb{E}[\sigma_t^2] = \frac{\omega}{1 - \alpha - \beta}$

- ▣ Derivation: take expectations in the recursion, $\mathbb{E}\sigma_t^2 = \omega + (\alpha + \beta)\mathbb{E}\sigma_{t-1}^2$, and set $\mathbb{E}\sigma_t^2 = \mathbb{E}\sigma_{t-1}^2$

- ▣ **IGARCH** (integrated GARCH): $\alpha + \beta = 1$; the unconditional variance is infinite

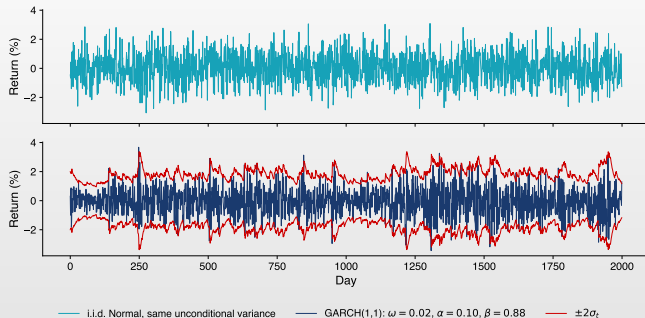
- ▶ Yet, if $\alpha > 0$ and z_t^2 is non-degenerate, the process is still **strictly** stationary: the condition $\mathbb{E} \ln(\alpha z_t^2 + \beta) < 0$ **Nelson (1990)** holds by Jensen, since $\mathbb{E} \ln(\alpha z_t^2 + \beta) < \ln(\alpha + \beta) = 0$

- ▣ Variance targeting: fix $\omega = \hat{\sigma}^2(1 - \alpha - \beta)$ with the sample variance $\hat{\sigma}^2$

Fourth Moment and Kurtosis

- With Normal z_t , the kurtosis of ε_t is $K = \frac{3[1 - (\alpha + \beta)^2]}{1 - (\alpha + \beta)^2 - 2\alpha^2}$
 - ▶ Finite only if $(\alpha + \beta)^2 + 2\alpha^2 < 1$ (Seminar 1, Exercise A9)
 - ▶ ARCH(1) is the case $\beta = 0$: $K = 3(1 - \alpha^2)/(1 - 3\alpha^2)$, finite only if $\alpha < 1/\sqrt{3}$: heavy tails even with Normal innovations
- S&P 500, GARCH(1,1) with Normal innovations: $\hat{\alpha} = 0.120604$, $\hat{\beta} = 0.859878$
 - ▶ $(\hat{\alpha} + \hat{\beta})^2 + 2\hat{\alpha}^2 = 0.990435$: just below 1, so $K = 12.1$; sample kurtosis of the returns: 13.7
- K is ill-conditioned near the moment boundary: report it from unrounded estimates (a rounding example: Appendix; Seminar 5, A4)
- Clustering alone generates much of the fat tails; the rest comes from heavy-tailed z_t

Simulation: GARCH Produces Clustering



- Top: i.i.d. Normal returns; bottom: GARCH(1,1) with $\omega = 0.02$, $\alpha = 0.10$, $\beta = 0.88$, the same unconditional variance (1), and the bands $\pm 2\sigma_t$
- Quiet stretches and bursts appear only in the GARCH path
- Normal innovations, yet fat-tailed returns: a mixture of Normal distributions with varying variances has heavy tails (kurtosis on a long path: Appendix)

Persistence and Half-Life

- Multi-step forecast: $\mathbb{E}[\sigma_{t+h}^2 \mid \mathcal{F}_t] = \sigma^2 + (\alpha + \beta)^{h-1}(\sigma_{t+1}^2 - \sigma^2)$
 - ▶ The gap to the long-run variance shrinks by the factor $\alpha + \beta$ each day
- **Half-life:** number of days after which half of a variance shock has gone
 - ▶ $(\alpha + \beta)^h = 0.5 \Rightarrow h_{1/2} = \frac{\ln 0.5}{\ln(\alpha + \beta)}$
- Examples: $\alpha + \beta = 0.95$: 13.5 days; 0.99: 69 days; 0.995: 138 days
- Small changes in $\alpha + \beta$ near 1 imply very different half-lives: estimation uncertainty matters (next section; Seminar 5, A3 and B7)

EWMA and IGARCH: No Mean Reversion

- ▣ **EWMA** J.P. Morgan/Reuters (1996): $\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2$, with $\lambda = 0.94$ for daily data and zero-mean returns
 - ▶ Weights $(1 - \lambda)\lambda^{j-1}$ on past squared returns: the most recent days count most
 - ▶ EWMA is an IGARCH(1,1) with $\omega = 0$, $\alpha = 1 - \lambda$, $\beta = \lambda$: no parameter to estimate
- ▣ Multi-step forecasts at the close of day t differ
 - ▶ EWMA: flat, $\mathbb{E}_t \sigma_{t+h}^2 = \sigma_{t+1}^2$ for every h
 - ▶ IGARCH with $\omega > 0$: rising, $\mathbb{E}_t \sigma_{t+h}^2 = \sigma_{t+1}^2 + (h - 1)\omega$
 - ▶ Stationary GARCH: pulled towards σ^2 at the rate $\alpha + \beta$
- ▣ EWMA serves as a benchmark in the forecast comparison

Maximum Likelihood

- Parameters $\theta = (\mu, \omega, \alpha, \beta)$; given θ , the recursion produces $\sigma_1^2(\theta), \dots, \sigma_n^2(\theta)$
- Normal log-likelihood: $\ell(\theta) = -\frac{1}{2} \sum_{t=1}^n [\ln 2\pi + \ln \sigma_t^2 + \varepsilon_t^2 / \sigma_t^2]$
 - ▶ Each day contributes the density of ε_t given the past: the likelihood factorises in time
- No closed form: numerical optimisation under $\omega > 0, \alpha, \beta \geq 0, \alpha + \beta < 1$
- Start-up: σ_1^2 is set to the sample variance (its effect fades at rate β)
- Scale: returns in %, so that the variance is of order 1
 - ▶ EUR/RON (daily s.d. about 0.28%) is fitted on 10 r_t : unscaled, the optimiser stopped at a local optimum

Gaussian QMLE: the Result and Its Assumptions

- **Gaussian QMLE:** maximise the Normal likelihood even if z_t is not Normal [Bollerslev & Wooldridge \(1992\)](#)
- **Consistency** (GARCH(1,1)) [Berkas, Horváth & Kokoszka \(2003\)](#), [Francq & Zakoïan \(2004\)](#)
 - ▶ Strict stationarity $\mathbb{E} \ln(\alpha z_t^2 + \beta) < 0$; θ_0 in a compact set; z_t^2 non-degenerate; no moment condition on ε_t
- **Asymptotic normality** of $\vartheta = (\omega, \alpha, \beta)$ (pure GARCH, known mean): add ϑ_0 interior and $\kappa_z = \mathbb{E} z_t^4 < \infty$
 - ▶ $\sqrt{n}(\hat{\vartheta} - \vartheta_0) \rightarrow N(0, (\kappa_z - 1)J^{-1})$, $J = \mathbb{E}[g_t g_t']$, $g_t = \partial \ln \sigma_t^2 / \partial \vartheta$
- Sandwich $\hat{A}^{-1} \hat{B} \hat{A}^{-1}$ estimates $(\kappa_z - 1)J^{-1}$; the inverse Hessian $\hat{A}^{-1}$ estimates $2J^{-1}$
 - ▶ A : minus the expected Hessian; B : variance of the score; robust / classical s.e. = $\sqrt{(\kappa_z - 1)/2}$, 1 under Normality
- Variants (estimated mean, IGARCH, explosive case): Appendix

QMLE in Our Data: When the Theory Holds and When It Fails

- S&P 500, GARCH(1,1) with Normal likelihood: residual kurtosis $\hat{\kappa}_z = 4.73$
 - ▶ Theory: robust / classical s.e. = $\sqrt{(4.73 - 1)/2} = 1.37$; observed: 1.37 for $\hat{\alpha}$, 1.37 for $\hat{\beta}$
- Bitcoin: GARCH-t gives $\hat{\nu} = 3.18 < 4$: under that law $\mathbb{E}z_t^4 = \infty$, and the condition for Gaussian-QMLE normality fails
 - ▶ Residual kurtosis does not settle: 19.8 on the first half, 14.6 on the full sample; s.e. ratios 2.51 and 2.01
- Hall & Yao (2003): then the Gaussian QMLE converges slower than $\sqrt{n}$, to a non-Gaussian limit: its robust t -statistics are not valid
- Remedies: a heavy-tailed quasi-likelihood with its scale adjustment Fan, Qi & Xiu (2014), or the Student-t MLE (bounded score: no fourth moment needed if the law is right), with bootstrap inference near bounds

Heavy-Tailed Innovations

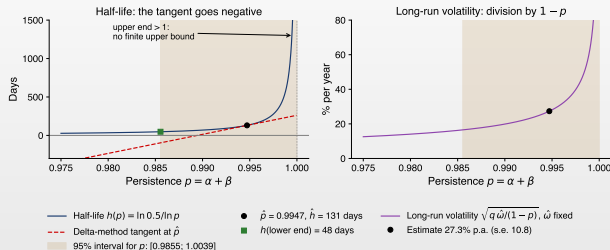
- **Student-t** [Bollerslev \(1987\)](#): $z_t = \sqrt{(\nu - 2)/\nu} t_\nu$, unit variance for $\nu > 2$
 - ▶ ν (degrees of freedom): the smaller ν , the heavier the tails; the fourth moment exists only for $\nu > 4$
- **Skewed-t** [Hansen \(1994\)](#): adds an asymmetry parameter $\lambda \in (-1, 1)$
 - ▶ $\lambda < 0$: the left tail is longer (large losses more likely than large gains)
- **GED** [Nelson \(1991\)](#): density $\propto \exp(-\frac{1}{2}|z/\lambda_\nu|^\nu)$
 - ▶ $\nu = 2$: the Normal distribution; $\nu = 1$: Laplace; $\nu < 2$: heavier tails (λ_ν gives unit variance)

S&P 500: GARCH(1,1)-t Estimates

Parameter	Estimate	Robust s.e.	Classical s.e.	Ratio
ω	0.0157			
α	0.123	0.011	0.010	1.06
β	0.872	0.010	0.010	1.08
ν	6.26	0.48		
$\alpha + \beta$	0.9947			

- Daily returns in %, 2000–2026; mean $\hat{\mu} = 0.079\%$ per day; full table with every standard error: Seminar 5, B1
- $\hat{\beta}$: most of today's variance is yesterday's variance; $\hat{\alpha}$: news adds a moderate update; $\hat{\nu} = 6.26$: tails clearly heavier than those of the Normal distribution
- Half-life 131 days; long-run volatility $\sqrt{251.4 \hat{\omega} / (1 - \hat{\alpha} - \hat{\beta})} = 27.3\%$ p.a., above the sample 19.2%: dividing by $1 - \hat{\alpha} - \hat{\beta}$ makes it imprecise (next slide)

Near-Unit Persistence: What the Delta Method Can and Cannot Do



- Delta method: $\widehat{\text{s.e.}} g(\hat{\theta}) \approx \sqrt{\nabla g' \hat{V} \nabla g}$, $\hat{V}$ the robust covariance of $(\hat{\omega}, \hat{\alpha}, \hat{\beta})$; long-run volatility 27.3% p.a., s.e. 10.8 points
- Half-life: $\partial h / \partial p = 24789$ at $\hat{p} = 0.994712$, $\widehat{\text{s.e.}}(\hat{p}) = 0.0047$, so $\widehat{\text{s.e.}}(\hat{h}) \approx 116$ days: the tangent goes negative
- Transform the interval for p instead: [0.9855; 1.0039] gives $h \in [48; \infty)$; whether $p = 1$ is plausible needs a boundary test (next slides)

Testing on the Boundary: a Decision Table

Situation under H_0	Example	Law of $LR = 2(\ell_1 - \ell_0)$	5% critical value
Interior restriction	GJR vs GARCH, $\gamma = 0$ (with $\alpha > 0$)	χ_1^2	3.84
One parameter on the boundary, the others identified and interior	Normal vs Student-t ($1/\nu = 0$); IGARCH ($\alpha + \beta = 1$)	$\frac{1}{2}\chi_0^2 + \frac{1}{2}\chi_1^2$ Self & Liang (1987), Andrews (2001)	2.71
A nuisance parameter unidentified	No ARCH: $\alpha = 0$ leaves β unidentified	non-standard Andrews (2001)	simulate
Several bounds active at once	GJR with $\hat{\alpha} = 0$, test of $\gamma = 0$	non-standard	simulate

- Non-nested models: $\text{AIC} = -2\ell + 2k$, $\text{BIC} = -2\ell + k \ln n$ (k parameters); lower is better
- A Wald t for an estimate at the bound (e.g. $\hat{\alpha} = 0$) is not $N(0, 1)$; a parametric bootstrap from $\hat{\theta}$ is inconsistent when θ_0 is on the boundary Andrews (2000): simulate under H_0 instead

Boundary Tests on Our Data

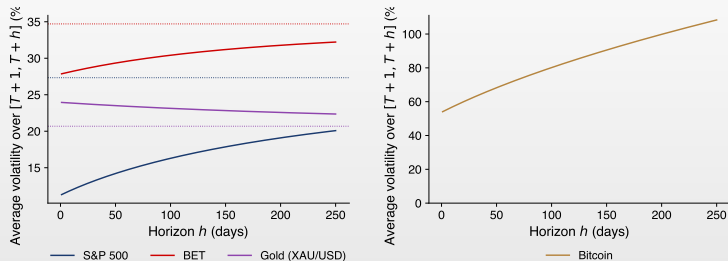
- Normal vs Student-t innovations, GARCH(1,1): $LR = 311.0$ (S&P 500), 1082.0 (Bitcoin)
 - ▶ Far above 2.71: heavy tails are decisive; the mixture matters only in borderline cases
- GJR-t, S&P 500: $\hat{\alpha} = 0.000$; its t -test is not standard, and the χ^2_1 law of the LR for $\gamma = 0$ assumes $\alpha_0 > 0$
- IGARCH, $H_0: \alpha + \beta = 1$ (GARCH-t): $LR = 1.11$ (S&P 500), 1.14 (BET)
 - ▶ S&P 500: $p = 0.29$ (χ^2_1), 0.15 (mixture), 0.17 (parametric bootstrap under H_0 , $B = 999$)
 - ▶ BET: $p = 0.28, 0.14$ and 0.17
- Under H_0 , 52% of the re-estimates sit at $\alpha + \beta = 1$, as the mixture predicts: a unit root in variance cannot be rejected for either index

Worked Example: One-Day and Ten-Day Forecasts

- S&P 500, GARCH(1,1)-t: $\hat{\omega} = 0.0157$, $\hat{\alpha} = 0.123$, $\hat{\beta} = 0.872$, $\hat{\mu} = 0.079$
- Last day $T = 18.09.2026$: $\hat{\varepsilon}_T = r_T - \hat{\mu} = 0.088$, filtered variance $\hat{\sigma}_T^2 = 0.565$
 - ▶ $\hat{\sigma}_{T+1}^2 = \hat{\omega} + \hat{\alpha}\hat{\varepsilon}_T^2 + \hat{\beta}\hat{\sigma}_T^2 = \hat{\omega} + 0.0009 + 0.493 = 0.509$, i.e. 11.3% p.a.
- Long-run variance $\hat{\sigma}^2 = \hat{\omega}/(1 - \hat{\alpha} - \hat{\beta}) = 2.97$; day 10: $\hat{\sigma}^2 + (\hat{\alpha} + \hat{\beta})^9(\hat{\sigma}_{T+1}^2 - \hat{\sigma}^2)$
 - ▶ $= 2.97 + 0.953 \times (0.509 - 2.97) = 0.624$
- 10-day variance $\sum_{h=1}^{10} \hat{\sigma}_{T+h}^2 = 5.67$, volatility 2.38%
 - ▶ The square-root-of-time rule gives $\sqrt{10} \times 0.714 = 2.26\%$: too low, because $\hat{\sigma}_{T+1}^2 < \hat{\sigma}^2$ and volatility is forecast to rise

 MFM_ch5_garch_estimation

Volatility Term Structure



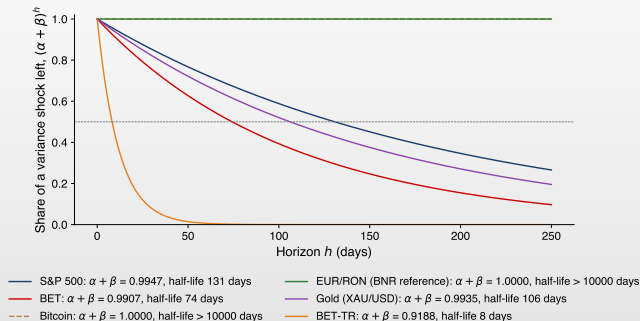
- Average volatility over the next h days, forecast from 18.09.2026 (GARCH(1,1)-t); dotted: long-run level
- S&P 500 rises from 11.3% to 20.1% in a year (below its long run today); gold falls from 23.9% to 22.4%
- Bitcoin, EUR/RON: IGARCH with $\hat{\omega} > 0$, the variance forecast grows by $\hat{\omega}$ per day (Bitcoin: 108.3% in a year): an artefact of the unit-root estimate

GARCH(1,1)-t in Six Markets

Series	$\hat{\omega}$	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\alpha} + \hat{\beta}$	$h_{1/2}$ (days)	$\hat{\nu}$	Long-run vol.
S&P 500	0.0157	0.123	0.872	0.9947	131	6.26	27.3
BET	0.0448	0.192	0.799	0.9907	74	5.01	34.7
BET-TR	0.0755	0.206	0.713	0.9188	8	5.02	15.2
Bitcoin	0.1937	0.109	0.891	1.0000	∞	3.18	–
EUR/RON	0.0000	0.123	0.877	1.0000	∞	4.20	–
Gold	0.0107	0.049	0.944	0.9935	106	4.58	20.7

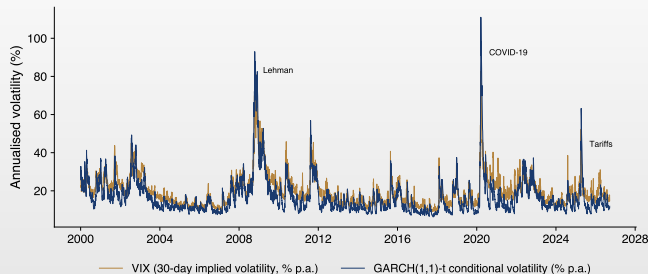
- Long-run volatility in % p.a., each series annualised with its own frequency; – when $\hat{\alpha} + \hat{\beta} \geq 0.999$; EUR/RON $\hat{\omega}$ is 0.0000 at four decimals
- Equities and gold: persistence 0.99–0.995; the BET reacts more to news ($\hat{\alpha} = 0.192$) and forgets faster than the S&P 500
- Different samples: BET-TR starts in 2014, so its lower persistence is not comparable with the BET 2000–2026; Bitcoin: $\hat{\nu} < 4$, an infinite fourth moment under the fitted law
- EUR/RON, a managed rate: on 6 May 2025 a move of +1.20% after days of $\pm 0.004\%$ gives a standardised residual of 154

Persistence: How Fast Do Shocks Fade?



- $(\hat{\alpha} + \hat{\beta})^h$: the share of a variance shock left after h days; dashed line: one half
- Half-lives: S&P 500 131 days, gold 106, BET 74, BET-TR 8; in Bitcoin and EUR/RON shocks never fade in the model
- Level shifts in volatility can mimic high persistence [Lamoureux & Lastrapes \(1990\)](#) (regime switching: [Hamilton & Susmel \(1994\)](#)): report half-lives as intervals (Seminar 5, B7)

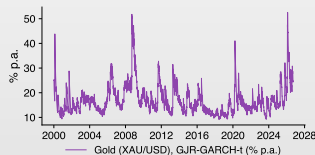
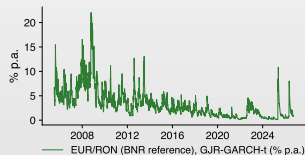
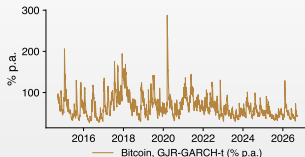
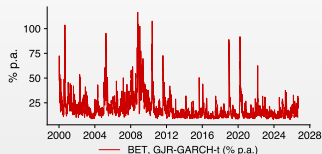
S&P 500: GARCH Volatility and the VIX



- $\hat{\sigma}_t \sqrt{252}$ from GARCH(1,1)-t and the VIX (30-day implied volatility), 2000–2026: correlation 0.89
- The VIX is higher on average: 19.8% against 17.0%, above GARCH on 84% of the days
- Not a measured variance risk premium: the premium compares VIX^2 with a 30-day variance forecast under the real-world probability; $\hat{\sigma}_t$ is a one-day volatility

 MFM_ch5_sp500_vix

BET, Bitcoin, EUR/RON and Gold



- GJR-GARCH-t volatility, each series annualised with its own frequency (Bitcoin: 365 days)
- BET: 2008 dominates, now 26.4%; Bitcoin: 50–100% is normal, the highest peak is March 2020
- EUR/RON: a managed calm with bursts in 2022 and 2025, poorly described by a daily GARCH; gold: a new high of 53% in 2026

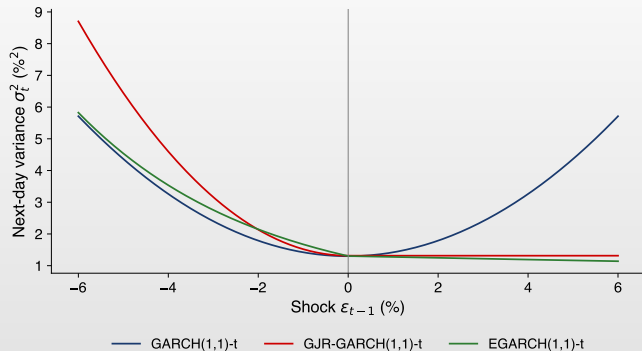
GJR-GARCH (Glosten, Jagannathan and Runkle, 1993)

- Bad news raises future volatility more than good news of the same size (stylised fact 9, Chapter 1); GARCH(1,1) cannot see the sign: ε_{t-1}^2 is the same for -3 and $+3$
- $\sigma_t^2 = \omega + (\alpha + \gamma I_{t-1}) \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$, with $I_{t-1} = 1$ if $\varepsilon_{t-1} < 0$ Glosten, Jagannathan & Runkle (1993)
 - ▶ Good news: slope α ; bad news: slope $\alpha + \gamma$; positivity: $\omega > 0$, $\alpha \geq 0$, $\alpha + \gamma \geq 0$, $\beta \geq 0$ (γ may be negative)
 - ▶ Persistence $\alpha + \gamma/2 + \beta$ for symmetric z_t ; for skewed z_t replace $1/2$ by $\mathbb{E}[z_t^2 1(z_t < 0)]$ (Seminar 5, A2)
- S&P 500: $\hat{\alpha} = 0.000$, $\hat{\gamma} = 0.205$ ($t = 10.0$), $\hat{\beta} = 0.881$; symmetry test: $H_0: \gamma = 0$ by t or by LR against GARCH(1,1)
- Candidate mechanisms, not tested here: leverage (a falling price raises the debt-to-equity ratio) and volatility feedback (higher expected volatility raises required returns, so prices fall now)

EGARCH (Nelson, 1991)

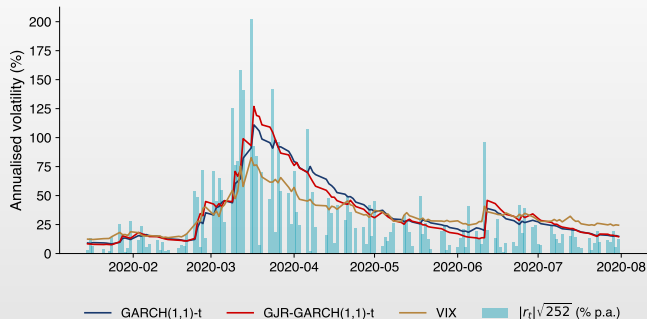
- $\ln \sigma_t^2 = \omega + \alpha (|z_{t-1}| - \mathbb{E}|z_{t-1}|) + \gamma z_{t-1} + \beta \ln \sigma_{t-1}^2$ Nelson (1991)
 - ▶ Size effect α , sign effect γ ; $\gamma < 0$: negative shocks raise volatility more
- The log form keeps $\sigma_t^2 > 0$ without any sign restriction on the parameters
- Persistence: β (stationary if $|\beta| < 1$)
- S&P 500: $\hat{\alpha} = 0.138$, $\hat{\gamma} = -0.165$ ($t = -15.2$), $\hat{\beta} = 0.979$
- In arch, $\mathbb{E}|z|$ is set to $\sqrt{2/\pi}$, its value under the Normal distribution

The News Impact Curve for the S&P 500



- News impact curve Engle & Ng (1993): σ_t^2 as a function of ε_{t-1} , with σ_{t-1}^2 fixed at the sample variance 1.47 (%²)
- Shock of -3% : GJR 3.16, EGARCH 2.76; $+3\%$: GJR 1.31, EGARCH 1.22; GARCH: 2.40 for both
- After bad news the forecast is 2.4 times (GJR) and 2.3 times (EGARCH) that after good news: ratios of levels, not of increases

Zoom: The COVID-19 Crash, 2020



- January–July 2020: $|r_t|$ annualised, GARCH-t and GJR-GARCH-t volatility, VIX
- 24 February ($r_t = -3.41\%$): the VIX went from 17.1% to 25.0% the same day; GARCH-t from 12.2% to 22.5% only the next day: GARCH reacts, markets price at once
- After $r_t = -12.8\%$ on 16 March, peaks on 17 March: GARCH-t 111%, GJR-t 127%, higher because the shocks were negative

Asymmetry in Six Markets

Series	$\hat{\gamma}_{GJR}$	t	LR vs GARCH	p-value	$\hat{\gamma}_{EGARCH}$	t
S&P 500	0.205	10.0	235.7	< 0.001	-0.165	-15.2
BET	0.043	2.1	4.9	0.026	-0.028	-2.6
BET-TR	0.111	2.9	10.3	0.001	-0.063	-3.5
Bitcoin	-0.013	-0.7	0.7	0.416	0.015	1.2
EUR/RON	0.039	3.0	9.3	0.002	-0.040	-2.5
Gold	-0.043	-4.7	27.6	< 0.001	0.038	5.7

- Student-t innovations; robust t; $LR \sim \chi^2_1$ (3.84) if the Student-t density is correct; the robust t is $N(0, 1)$ only if the other parameters are interior (not for $\hat{\alpha} = 0$)
- Bad news weighs more (GJR $\hat{\gamma} > 0$, EGARCH $\hat{\gamma} < 0$): S&P 500 strongly ($LR = 235.7$), BET and BET-TR weakly (4.9, 10.3), EUR/RON slightly; Bitcoin: none detected ($LR = 0.7$); gold: inverted ($\hat{\gamma}_{GJR} = -0.043$)
- Leverage for equities and a safe-haven role for gold are interpretations consistent with these signs, not tested mechanisms

S&P 500: Twelve Models Compared by BIC

ΔBIC	$N(0, 1)$	Student-t	Skewed-t	GED
GARCH(1,1)	648.9	346.7	307.4	339.8
GJR-GARCH(1,1)	394.7	119.9	49.6	141.6
EGARCH(1,1)	372.9	74.8	0.0	111.4

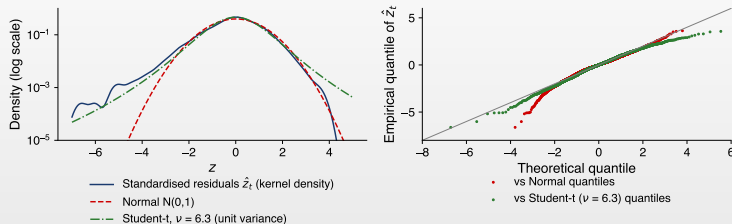
- ΔBIC : $\text{BIC} = -2\ell + k \ln n$ minus the lowest BIC among the 12 models (0 = preferred model)
- Two choices matter: Student-t instead of Normal innovations lowers BIC by about 300; asymmetry lowers it by a further 230–270; winner: EGARCH with skewed-t innovations
- BIC ranks in-sample fit, not forecast accuracy: the forecast section tests whether these gains survive out of sample

 MFM_ch5_garch_estimation

Standardised Residuals and the Sign-Bias Tests

- **Standardised residual:** $\hat{z}_t = \hat{\varepsilon}_t / \hat{\sigma}_t$; if the model is right, $\hat{z}_t$ is close to i.i.d. with variance 1
- Checks, each with H_0 : the model captures that feature
 - ▶ Mean equation: Ljung–Box $Q(10)$ on $\hat{z}_t$; variance equation: $Q(10)$ on $\hat{z}_t^2$ and LM(5); distribution: QQ plot against the assumed density
- Sign-bias regression [Engle & Ng \(1993\)](#): $\hat{z}_t^2 = b_0 + b_1 S_{t-1}^- + b_2 S_{t-1}^- \hat{\varepsilon}_{t-1} + b_3 S_{t-1}^+ \hat{\varepsilon}_{t-1} + u_t$
 - ▶ $S_{t-1}^- = 1$ if $\hat{\varepsilon}_{t-1} < 0$, else 0; $S_{t-1}^+ = 1 - S_{t-1}^-$; the size terms use the unstandardised residual $\hat{\varepsilon}_{t-1}$ (eq. 18)
 - ▶ b_1 : sign bias; b_2 , b_3 : negative and positive size bias; joint test $nR^2 \sim \chi_3^2$ (7.81 at 5%), conservative because it omits the gradient of the fitted variance

S&P 500: Are the Innovations Normal?



- Standardised residuals of GARCH(1,1) with Normal innovations: excess kurtosis falls from 10.7 (returns) to 1.73; skewness -0.53, Jarque-Bera 1153
- 15 residuals below -4: 0.21 expected under the Normal, 8.5 under a Student-t with $\nu = 6.3$; largest: $\hat{z} = -6.6$ (27 February 2007)
- A longer left tail: skewed-t with $\hat{\lambda} = -0.150$ (GJR, $\hat{\eta} = 7.7$), hence BIC prefers skewed-t innovations

S&P 500: Diagnostics of Five Models

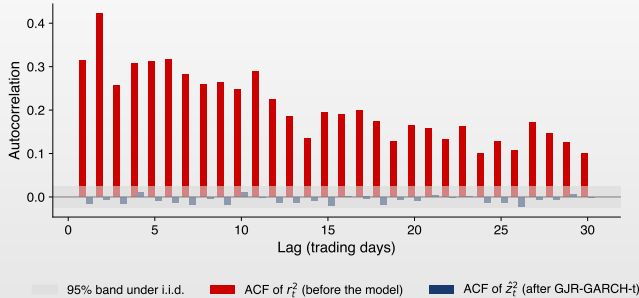
Model	Q(10) on $\hat{z}$	Q(10) on $\hat{z}^2$	LM(5)	Sign bias t	Joint χ^2_3
GARCH-N	21.2	16.4	8.0	3.64	41.8
GARCH-t	21.9	13.6	5.8	3.56	40.7
GJR-t	22.3	10.2	4.5	3.61	28.1
EGARCH-t	27.6	10.5	8.2	2.04	9.9
GJR-skewed-t	19.7	10.7	5.0	3.91	31.8

- 5% critical values: 18.3 (χ^2_{10}), 11.1 (χ^2_5), 7.81 (χ^2_3), $|t| > 1.96$; naive laws, conservative with estimated parameters (next slide)
- Mean: Q(10) on $\hat{z}$ of 20–28 in every model: a little autocorrelation that a constant mean ignores
- Variance: Q(10) on $\hat{z}^2$ falls from 6133 (on r_t^2) to 10–16 and LM(5) is not significant: these naive tests detect no clustering left
- Asymmetry: symmetric GARCH fails the joint test (40.7), GJR still fails (28.1), EGARCH comes closest (9.9, $p = 0.020$); with $n = 6\,714$, judge economic size, not only p-values

Residual Diagnostics Have Estimated Parameters

- ▣ $\hat{z}_t$ depends on $\hat{\theta}$: the Ljung–Box statistic on $\hat{z}_t^2$ is not exactly χ_m^2 Li & Mak (1994); the fitted variance has already absorbed part of the dependence at small lags
- ▣ S&P 500, Gaussian QMLE of GARCH(1,1), $Q(10)$ on $\hat{z}_t^2$
 - ▶ Naive Ljung–Box: 16.4, $p = 0.089$: not rejected at 5%
 - ▶ Corrected for the estimated parameters: 19.1, $p = 0.038$: rejected at 5%, the verdict flips
- ▣ The naive test is conservative: it rejects a misspecified variance equation too rarely; the joint sign-bias test carries the same caveat
- ▣ The correction matrix and its entries: Appendix

Diagnostics: Squared Returns versus Squared Residuals



- S&P 500, lags 1–30: autocorrelations of r_t^2 (before the model) and of $\hat{z}_t^2$ from GJR-GARCH-t
- After the model they sit inside the i.i.d. band: no remaining variance dynamics are detected at these lags
- The band is the naive one: with estimated parameters it is too wide at small lags (previous slide)

Diagnostics in Other Markets: a Decision Table

Market	Residual pattern	Possible failure	Next check
BET	$Q(10)$ on $\hat{z} = 116.6$	autocorrelation in the mean (thin trading, Chapter 2)	AR(1) mean equation
Bitcoin	$Q(10)$ on $\hat{z}^2 = 4.7$, joint sign bias 1.3: no rejection	$\hat{\nu} = 3.18 < 4$: $\text{Var}(z_t^2) = \infty$, so the χ^2 calibration is not justified	tests on $ \hat{z}_t $
EUR/RON	excess kurtosis 3190	one residual of 154 (May 2025) dominates	inspect the largest residuals; a bounded-score model
Gold	EGARCH: $Q(10)$ on $\hat{z}^2 = 48.8$	clustering left despite a good BIC	choose by diagnostics, not by BIC alone

- Always look at the largest residuals: one observation can dominate every diagnostic

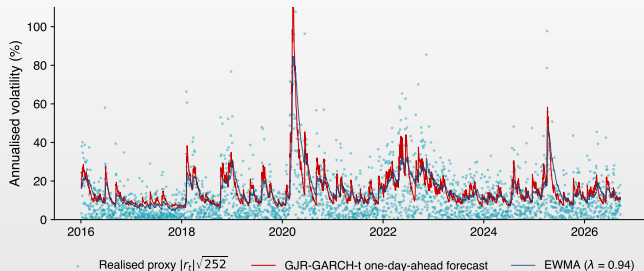
The Proxy Problem and Robust Losses

- Volatility is never observed: we compare forecasts with a noisy **proxy**, e.g. r_t^2
- $r_t^2 = \sigma_t^2 z_t^2$ is unbiased ($\mathbb{E}[r_t^2 \mid \mathcal{F}_{t-1}] = \sigma_t^2$ for $\mu = 0$) but very noisy
 - ▶ Its R^2 against the true variance is small even for a perfect forecast [Andersen & Bollerslev \(1998\)](#)
 - ▶ S&P 500: we ignore the mean, since $\hat{\mu}^2 = 0.0062$ against an average daily variance of 1.47 ($\%^2$), a bias below 1%
- [Patton \(2011\)](#): only some loss functions rank forecasts correctly with a noisy proxy
 - ▶ Robust: MSE $(r_t^2 - h_t)^2$ and **QLIKE** $r_t^2/h_t + \ln h_t$, with h_t the variance forecast
 - ▶ Not robust: losses on volatility $|r_t| - \sqrt{h_t}$ or on logs
- QLIKE is (up to constants) minus the Normal log-likelihood; it penalises under-prediction more

Mincer–Zarnowitz and Diebold–Mariano

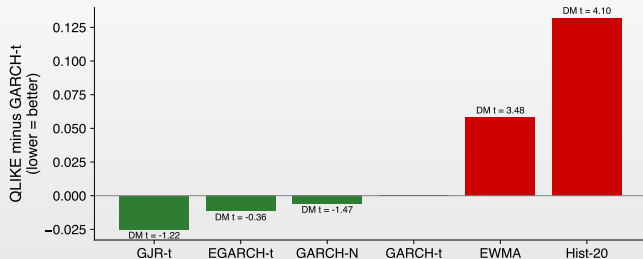
- **Mincer–Zarnowitz regression** Mincer & Zarnowitz (1969): $r_t^2 = a + b h_t + u_t$
 - ▶ Unbiased forecast: $a = 0$, $b = 1$ (joint Wald test with HAC errors); R^2 measures information content
 - ▶ $b < 1$: forecasts too dispersed (too high when high, too low when low)
- **Diebold–Mariano test** Diebold & Mariano (1995): loss differential $d_t = L(r_t^2, h_t^A) - L(r_t^2, h_t^B)$
 - ▶ $DM = \bar{d} / \widehat{\text{s.e.}}(\bar{d})$ with a HAC standard error; $\approx N(0, 1)$ under equal accuracy; $DM < -1.96$: A significantly better
 - ▶ The $N(0, 1)$ law assumes given forecasts; for estimated, nested models (GARCH-t inside GJR-t) it can fail, so DM t is read descriptively there (GW slide below)

Out-of-Sample Design and Forecast Paths, 2016–2026



- S&P 500: 2 692 one-day forecasts; refit each January on all data since 2000 (expanding window), daily updates with fixed parameters
- Timing: h_t uses data up to the close of day $t - 1$ and is compared with r_t^2 ; changing r_t must leave h_t unchanged
- Models: GARCH-N, GARCH-t, GJR-t, EGARCH-t; benchmarks: EWMA and Hist-20 (20-day historical variance); shown: GJR-t, EWMA, $|r_t|\sqrt{252}$

S&P 500: Does Anything Beat GARCH(1,1)?



- Mean QLIKE minus that of GARCH-t, with the DM t (HAC, 5 lags); lowest QLIKE: GJR-t (0.732)
- GARCH family: DM t from -1.47 to -0.36, descriptive for these nested pairs
- EWMA and Hist-20 are clearly worse ($t = 3.48, 4.10$): mean reversion and estimated parameters pay off [Hansen & Lunde \(2005\)](#)
- MZ: unbiasedness not rejected for GARCH-type forecasts, rejected for Hist-20 (all statistics: Appendix)

 MFM_ch5_forecast_evaluation

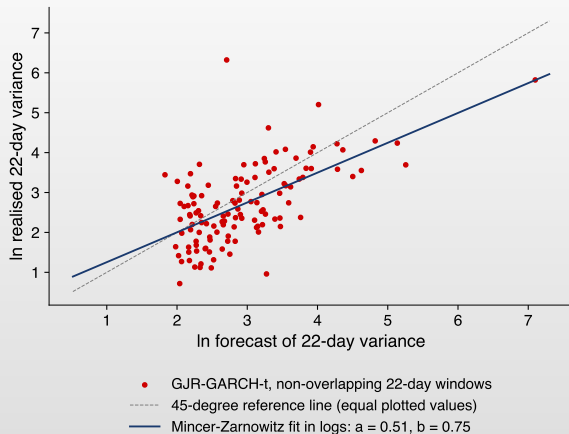
Comparing Forecasting Methods: Giacomini–White and the Model Confidence Set

- With estimated parameters, nested models and an expanding window, the DM differential degenerates under H_0 : DM is not $N(0, 1)$ West (1996), Clark & McCracken (2001)
- Giacomini & White (2006): test the forecasting **method** (model + rolling window of $W = 1\,000$ days), valid for nested models because the estimation noise does not vanish
- Six models, five pairwise tests: a multiple-comparison problem; **Model Confidence Set** Hansen, Lunde & Nason (2011) drops the worst model until equal accuracy is not rejected; the set contains the best model with probability $\geq 1 - \alpha$
- Result: GW rejects only EWMA and Hist-20; the 90% MCS keeps the four GARCH models: the DM verdict, now calibrated (formulas and full table: Appendix)

Model	GW p	MCS p
GARCH-N	0.054	0.586
GARCH-t	–	0.353
GJR-t	0.288	1.000
EGARCH-t	0.201	0.586
EWMA	0.003	0.008
Hist-20	< 0.001	0.001

- GW: rolling window, 2 692 forecasts, vs GARCH-t; MCS: expanding window, QLIKE, stationary bootstrap

One-Month Forecasts: Mincer–Zarnowitz in Logs



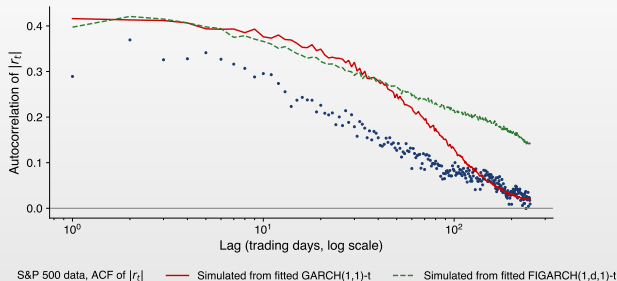
- GJR-t forecast of the 22-day variance against the realised sum of 22 squared returns, 122 non-overlapping windows; R^2 in logs 0.38
- Slope in logs $b = 0.75$ (s.e. 0.07): descriptive only, since $\mathbb{E}[\ln Y | \mathcal{F}] \neq \ln \mathbb{E}[Y | \mathcal{F}]$; $b < 1$ does not prove over-dispersed forecasts
- In levels b is only 0.27: two COVID-19 windows dominate; compare forecasts with QLIKE [Patton \(2011\)](#)

GARCH-in-Mean: Is Risk Rewarded?

- **GARCH-M** Engle, Lilien & Robins (1987): $r_t = c + \kappa \sigma_t + \varepsilon_t$ (or $\kappa \sigma_t^2$)
 - ▶ $\kappa > 0$: investors demand a higher expected return when risk is higher (risk-return trade-off)
- S&P 500, GARCH(1,1)-t in mean: $\hat{\kappa} = 0.096$ (t = 3.02) with σ_t ; 0.028 (t = 2.35) with σ_t^2
 - ▶ Positive and significant in this sample, but the literature is mixed: GJR (1993) found a negative relation
- Caution: a daily mean effect is small and fragile; it is sensitive to the sample and to the asymmetry in variance

 MFM_ch5_long_memory

Long Memory: Where GARCH Falls Short



- S&P 500, ACF of $|r_t|$ (log scale): data 0.289, 0.082, 0.009 at lags 1, 100, 250; GARCH(1,1)-t too high at lag 1 (0.416), then an exponential collapse
- **FIGARCH** Baillie, Bollerslev & Mikkelsen (1996), $(1 - L)^d$: hyperbolic decay; $\hat{d} = 0.60$ (s.e. 0.06), BIC lower by 34.2, but the memory is now too long (0.144 at lag 250)
- No single-scale model fits all horizons: realised volatility and HAR in Chapter 9

Limits of GARCH and the Extensions They Motivate

Limit	Evidence in this chapter	Extension
Level shifts inflate $\alpha + \beta$ Lamoureux & Lastrapes (1990)	BET-TR (2014–2026) 0.9188 vs BET (2000–2026) 0.9907	GARCH-MIDAS, a slow long-run component Engle, Ghysels & Sohn (2013) ; regime switching Hamilton & Susmel (1994)
Reaction lag of one day	COVID-19: the VIX moved the same day, GARCH the next	Realized GARCH, with a realised-variance measurement equation Hansen, Huang & Shek (2012) (Chapter 9)
One outlier drives the update	EUR/RON residual of 154 (May 2025)	Score-driven (GAS) models with a bounded score Creal, Koopman & Lucas (2013) ; the case study below
Short memory	ACF of $ r_t $ at lags 100–250	FIGARCH; HAR on realised variance (Chapter 9)
One asset at a time	portfolios need covariances	Multivariate GARCH and DCC (Chapter 6)
Managed prices	EUR/RON: long calm, then jumps	Model interventions as jumps or regimes

- Each extension is judged like GARCH: out of sample, robust loss, Model Confidence Set (model equations: Appendix)

Case Study: Question and Model

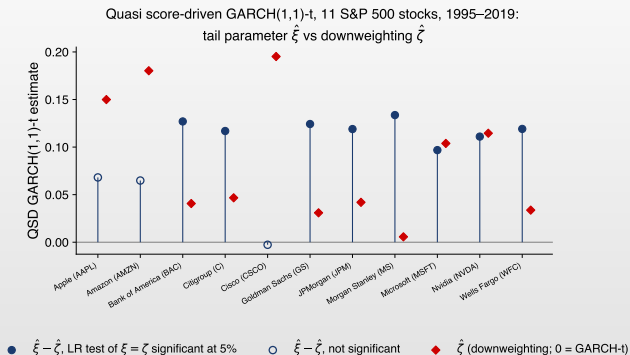
- **Question** Blasques, Francq & Laurent (2023): must a score-driven volatility model update σ_t^2 with the score of the same density that describes the returns?
- Score-driven (GAS) update Creal, Koopman & Lucas (2013): $f_{t+1} = \omega + \alpha S(f_t) \partial \log p(y_t | f_t) / \partial f_t + \beta f_t$; with a Student-t density one ν sets both the tails of z_t and how strongly a large shock is downweighted
- **QSD GARCH(1,1)-t**, eq. (30): $y_t = \sqrt{f_t} \epsilon_t$, ϵ_t standardised Student-t with $1/\xi$ degrees of freedom (f_t is σ_t^2 , ϵ_t is z_t)
 - ▶ $f_{t+1} = \omega + \varpi X_t + \alpha \frac{1 + \zeta}{1 - 2\zeta + \zeta \epsilon_t^2} \epsilon_t^2 f_t + \beta f_t$
 - ▶ Two parameters: ξ , the tail thickness of ϵ_t , and ζ , the downweighting of large ϵ_t^2 in the update; $\zeta = 0$: GARCH(1,1)-t; $\zeta = \xi$: Beta-t-GARCH (score-driven)
- Data: 400 S&P 500 stocks, 3 January 1995 (or later) – 28 February 2019; X_t : the squared VIX at a daily horizon; maximum likelihood with an AR(1) mean; LR tests of $\zeta = 0$ and $\xi = \zeta$ against χ_1^2 (full method: Appendix)

Case Study: Key Results (Tables 4–6)

Rejected at 5%, % of the 400 stocks	Student-t (Table 4)	Skewed t (Table 6)
LR, $H_0: \zeta = 0$ (GARCH update)	89.5	98.8
LR, $H_0: \xi = \zeta$ (score-driven; skewed: also $\kappa = \tau$)	53.5	100.0
PIT test: GARCH(1,1)	40.0	33.1
PIT test: Beta-t-GARCH	46.0	72.2
PIT test: QSD GARCH(1,1)	45.7	11.3

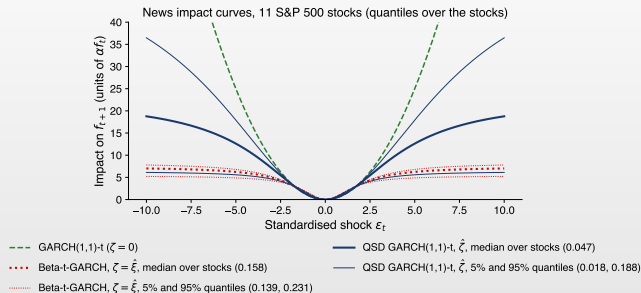
- Downweighting large shocks matters: $\zeta = 0$ is rejected for 89.5% of the stocks and every $\hat{\zeta}$ is positive (Table 4)
- But Beta-t-GARCH downweights too much: where $\hat{\xi} - \hat{\zeta}$ is significant, $\hat{\xi} > \hat{\zeta}$ for all stocks but one; median $1/\hat{\zeta}$ is 9.48, against 5.07 (Figs. 2–3): tails and downweighting are two different parameters
- Skewed t: $\hat{\tau}$ far below 0.5 (median 0.333) while $\hat{\kappa}$ stays near 0.5 (Figs. 5–6): leverage enters through the update, not the density; the QSD model then fails the PIT test for only 11.3% of the stocks

Case Study: Replication on 11 Stocks (Fig. 2)



- Course sample: 11 large stocks still listed (the paper: 400), 1995 (or later) – February 2019, 4986–6081 returns each;
 $X_t = \text{VIX}_t^2 / 252$
- $\zeta = 0$ rejected for 9 of 11 (paper: 89.5%);
 $\xi = \zeta$ rejected for 8 (paper: 53.5%); $\hat{\xi} > \hat{\zeta}$ for 10 of 11
- Median $1/\hat{\zeta}$: 21.4 (QSD) against 6.32 (Beta-t-GARCH); paper: 9.48 against 5.07;
11 selected survivors: imprecise and not representative

Case Study: News Impact Curves (Fig. 3)



- Weight of ϵ_t in f_{t+1} (units of αf_t) at quantiles of the 11 estimates: updating functions at a common αf_t , not fitted paths
- A 5-s.d. shock adds 25 (GARCH-t), 12.6 (median QSD), 6.2 (median Beta-t-GARCH); at 10 s.d.: 100, 18.8, 7.0 (bound $(1 + \zeta)/\zeta = 7.3$)
- QSD lies between the two: large shocks are downweighted, far less than Beta-t-GARCH imposes

An Open Question and the Discovery Loop

- Do machine-learning or realised-measure forecasts beat a well-specified GARCH for daily equity variance, out of sample, once the number of models compared is accounted for?
 - ▶ Christensen, Siggaard & Veliyev (2023): machine-learning methods with little tuning are competitive with HAR-type models; Hansen, Lunde & Nason (2011): the Model Confidence Set is the right object when many models compete
 - ▶ Formal: $d_t = L(r_t^2, h_t^A) - L(r_t^2, h_t^B)$, $H_0: \mathbb{E}d_t = 0$; falsified if A leaves the Model Confidence Set, or if the sign of $\bar{d}$ flips across pre-registered subperiods
- Loop: an AI assistant drafts the literature map, competing hypotheses, code, a robustness plan and a referee-style critique; the researcher resolves every DOI, reproduces every number and fixes the plan before the first run
- Checks specific to volatility forecasts
 - ▶ Look-ahead (h_t must not move when r_t changes); proxy-robust loss (QLIKE, MSE); model mining (Model Confidence Set or Bonferroni); scale and convergence of the optimiser; invented DOIs or arch arguments
 - ▶ Reusable prompts and the full checklist: Appendix

Mini-Case and a Team-Project Seed

- A typical assistant claim: “GJR-t beats GARCH-t because of the leverage effect”; $d_t = \text{QLIKE}(\text{GJR-t}) - \text{QLIKE}(\text{GARCH-t})$, 2 692 days, 2016–2026: $\bar{d} = -0.025$, DM $t = -1.22$
- Testable implication of leverage: the gain sits after down days
 - ▶ After $r_{t-1} < 0$ (1 223 days): $\bar{d} = -0.015$, $t = -0.51$; after $r_{t-1} \geq 0$ (1 469): $\bar{d} = -0.034$, $t = -1.12$: not supported; 2016–2020 $t = -0.02$, 2021–2026 $t = -4.06$
 - ▶ Descriptive only (nested models, expanding window): the calibrated follow-up is a fixed-window GW test of the sign difference, with d_t regressed on $1(r_{t-1} < 0)$
- **Project seed:** do machine-learning forecasts enter the Model Confidence Set together with GARCH, GJR and EGARCH for the S&P 500 and the BET, 2016–2026?
 - ▶ Replicate the QLIKE table first; pre-register models, losses, splits and error rate; log AI use in AI_USE.md and every caught error in AI_ERRORS.md (full brief: Appendix)

Synthesis: a Decision Workflow

- ▣ 1. **Detect:** $Q(10)$ on r_t^2 and the LM test; clustering is universal, persistence $\alpha + \beta$ is close to 1 in all our markets
- ▣ 2. **Fit:** GARCH, GJR, EGARCH with Normal, Student-t, skewed-t innovations; returns in %; check convergence and parameters at a bound
- ▣ 3. **Diagnose:** level, squares and signs of $\hat{z}_t$ and its largest values; naive tests are conservative
- ▣ 4. **Forecast:** the recursion gives multi-step variances; the square-root-of-time rule errs in the direction of mean reversion
- ▣ 5. **Compare:** out of sample, QLIKE, GW with a rolling window and the Model Confidence Set: GARCH beats EWMA and Hist-20, the GARCH family is hard to separate
- ▣ 6. **Report uncertainty:** robust s.e. (valid if $\mathbb{E}z_t^4 < \infty$), boundary-aware tests, half-lives as intervals $[h_{lo}; \infty)$ when $\alpha + \beta \approx 1$

Check Yourself, and Next: Chapter 6

Retrieval questions

- Why does the square-root-of-time rule overstate the 10-day risk after a turbulent day?
- Why is the LR test of Normal against Student-t innovations compared with 2.71 and not 3.84?
- Why can a symmetric interval for the half-life contain negative values?
- Why is a gap between the VIX and GARCH volatility not a measured variance risk premium?

Next chapters

- Chapter 6: multivariate volatility and dynamic correlations (DCC)
- Chapters 7–8: Value at Risk and expected shortfall with GARCH volatility, and their backtesting
- Chapter 9: realised volatility and long memory
- Seminar 5: the variance recursion step by step, GARCH-t and GJR on real data, an AI answer to audit, bootstrap tests and forecast comparisons

Appendix: Data Used in This Chapter

Series	Owner / source	Start	Obs.	Obs. per year
S&P 500	S&P Dow Jones Indices	2000	6 714	251.4
BET	Bucharest Stock Exchange	2000	6 670	249.8
BET-TR	Bucharest Stock Exchange	2014	2 992	249.7
Bitcoin (BTC/USD)	crypto exchanges (composite)	2014	4 384	365.3
EUR/RON reference rate	BNR	2005	5 352	252.4
Gold (XAU/USD)	spot market	2000	6 922	259.2

- Daily closes up to 18 September 2026; returns $r_t = 100 (\ln P_t - \ln P_{t-1})$, in %
- Equity indices and gold: weekdays only; gold volatility is annualised using its actual observation frequency
- Bitcoin trades 7 days a week; the VIX index of the Chicago Board Options Exchange is used as a benchmark

 MFM_ch5_arch_effects

Appendix: ARCH Effects, All Statistics

Series	Vol. (% p.a.)	Excess kurt.	$\hat{\rho}_1(r^2)$	$Q(10)$ on r^2	LM(5)
S&P 500	19.2	10.7	0.313	6133	1684
BET	22.1	11.1	0.330	2440	982
BET-TR	15.4	17.8	0.171	714	371
Bitcoin	66.6	12.0	0.137	213	115
EUR/RON	4.4	15.1	0.342	2420	871
Gold	17.6	7.7	0.110	744	329

□ 5% critical values: χ_{10}^2 : 18.3; χ_5^2 : 11.1; all p-values < 0.001

 MFM_ch5_arch_effects

Appendix: One-Sided Score Tests and the Davies Problem

- The alternative is one-sided ($\alpha_i \geq 0$): the two-sided χ^2 test wastes power
 - ▶ Lee & King (1993): one-sided score test built on the sum of the score components, $N(0, 1)$ under H_0 , rejects in the upper tail only
- Against GARCH(1,1) the null $\alpha = 0$ leaves β unidentified (Davies problem)
 - ▶ The ARCH(q) LM statistic is also the score test against GARCH(p, q) Lee & King (1993): the same statistic serves both alternatives

Appendix: Robert Engle and the 2003 Nobel Prize



Robert F. Engle (2022)

Photo: Jon Demske (2022); CC

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Clive W.J. Granger (2008)

Photo: Olaf Storbeck (2008); CC

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- Nobel Prize 2003: Engle "for methods of analyzing economic time series with time-varying volatility (ARCH)"
 - ▶ Shared with Clive Granger (cointegration)
- Nobel lecture: Engle (2004), on risk and volatility in financial practice
- Tim Bollerslev, Engle's doctoral student, generalised ARCH to GARCH in 1986

Appendix: Kurtosis: Rounding and a Long Simulated Path

- S&P 500, GARCH-N: the moment condition is 0.990435, so $K = 12.1$ has the denominator $1 - 0.990435$
 - ▶ Rounding to $\hat{\alpha} = 0.121$, $\hat{\beta} = 0.860$ gives 0.9916 and $K = 13.5$: report such functions from unrounded estimates
- Simulated GARCH(1,1) of the simulation slide, one million days
 - ▶ Excess kurtosis 2.47 (theory 3.06); lag-1 autocorrelation of ε_t^2 : 0.26
 - ▶ The sample kurtosis converges slowly: its precision depends on the eighth moment, infinite for these parameters

Appendix: QMLE: Variants of the Theorem

- ▣ Estimated mean: the score gains the term ε_t/σ_t^2 and the covariance $(\kappa_z - 1)J^{-1}$ changes form
- ▣ IGARCH ($\alpha + \beta = 1$): strict stationarity still holds, and the QMLE remains consistent and asymptotically Normal [Lumsdaine \(1996\)](#)
- ▣ Explosive case, $\mathbb{E} \ln(\alpha z_t^2 + \beta) \geq 0$: [Jensen & Rahbek \(2004\)](#) cover (α, β) with ω fixed, since ω is not identified
- ▣ Heavy tails, $\kappa_z = \infty$: slower than $\sqrt{n}$, non-Gaussian limits [Hall & Yao \(2003\)](#); heavy-tailed quasi-likelihoods with a scale adjustment [Fan, Qi & Xiu \(2014\)](#)

Appendix: The Li-Mak Correction for Squared-Residual Autocorrelations

- Squared-residual autocorrelations $\hat{r}_k$: $\sqrt{n} \hat{r} \rightarrow N(0, C)$, $C = I - H' J^{-1} H / (\kappa_z - 1)$ Li & Mak (1994)
 - ▶ $H_k = \mathbb{E}[g_t(z_{t-k}^2 - 1)]$: how the fitted variance absorbs lag- k dependence; the ARMA rule χ_{m-p-q}^2 does not apply
- S&P 500, Gaussian QMLE of GARCH(1,1): $C_{1,1} = 0.78$, $C_{5,5} = 0.93$, $C_{10,10} = 0.97$
 - ▶ The correction is largest at small lags, where $\hat{\alpha}$ has already fitted the dependence; corrected statistic $n\hat{r}'C^{-1}\hat{r} = 19.1$

 MFM_ch5_inference

Appendix: S&P 500 Forecast Evaluation, All Statistics

Model	QLIKE	MSE	DM t vs GARCH-t	MZ b	MZ R^2	Wald p
GARCH-N	0.751	22.3	-1.47	0.97	0.250	1.00
GARCH-t	0.757	22.8	—	0.88	0.238	0.80
GJR-t	0.732	21.8	-1.22	0.84	0.276	0.68
EGARCH-t	0.745	21.3	-0.36	1.24	0.293	0.71
EWMA	0.815	24.7	3.48	0.89	0.172	0.69
Hist-20	0.889	26.1	4.10	0.68	0.152	0.03

- Mean losses on 2 692 days; DM on QLIKE, HAC with 5 lags; negative t: better than GARCH-t
- MZ: $r_t^2 = a + b h_t$; Wald p: test of $a = 0$, $b = 1$

 MFM_ch5_forecast_evaluation

Appendix: Giacomini–White and the MCS: Formulas and Full Table

Model	QLIKE (W)	t uncond.	GW	p	MCS p
GARCH-N	0.751	1.87	5.85	0.054	0.586
GARCH-t	0.743	–	–	–	0.353
GJR-t	0.738	-0.18	2.49	0.288	1.000
EGARCH-t	0.746	0.07	3.21	0.201	0.586
EWMA	0.815	3.20	11.40	0.003	0.008
Hist-20	0.889	3.87	16.41	< 0.001	0.001

- Conditional test $\mathbb{E}[d_{t+1} \mid \mathcal{F}_t] = 0$ with instruments $h_t = (1, d_t)$: $GW = n \bar{Z}' \hat{\Omega}^{-1} \bar{Z} \rightarrow \chi^2_2$, $Z_{t+1} = h_t d_{t+1}$ (critical value 5.99)
- Rolling window $W = 1000$ days, re-estimated each January, 2692 forecasts; tests against GARCH-t, HAC with 5 lags
- MCS on the expanding-window forecasts: QLIKE, statistic T_R , stationary bootstrap (mean block 10, 10 000 draws); 90% MCS: GARCH-N, GARCH-t, GJR-t, EGARCH-t

Appendix: Score-Driven, Realized and Component Models

- **Score-driven (GAS)** Creal, Koopman & Lucas (2013): $\sigma_{t+1}^2 = \omega + A s_t + B \sigma_t^2$, s_t the scaled score of the Student-t density
 - ▶ The score is bounded in z_t : an outlier such as the EUR/RON residual of 154 moves the variance far less than in GARCH
- **Realized GARCH** Hansen, Huang & Shek (2012): adds a measurement equation linking σ_t^2 to a realised variance
 - ▶ Uses intraday information and reacts faster; the bridge to Chapter 9
- **GARCH-MIDAS** Engle, Ghysels & Sohn (2013): $\sigma_t^2 = \tau_t g_t$, a slow long-run component τ_t driven by macro data (mixed frequencies)
 - ▶ A moving long-run level absorbs the level shifts that push $\hat{\alpha} + \hat{\beta}$ towards 1

Appendix: Beyond Variance: Entropy and Tail Risk

- ▣ [Pele, Lazar & Dufour \(2017\)](#): entropy of the intraday return distribution and market risk (EUR/JPY)
 - ▶ Higher entropy goes with lower intraday Value at Risk and expected shortfall; entropy then forecasts daily Value at Risk
- ▣ [Pele & Mazurencu-Marinescu-Pele \(2019\)](#): the entropy of intraday Bitcoin returns proposed as an additional signal for daily Value at Risk
- ▣ [Pele, Lazar & Mazurencu-Marinescu-Pele \(2019\)](#): a nonparametric tail measure based on entropy, compared with standard expected shortfall models
- ▣ GARCH volatility feeds directly into Value at Risk and expected shortfall: Chapters 7–8

Appendix: Case Study, Full Method

- **Why it matters:** GARCH is driven by ε_t^2 whatever the density, so an isolated extreme shock can produce an excessive volatility response lasting days or weeks (Section 1 of the paper)
 - ▶ With a left-skewed density, a pure score-driven model raises volatility more after positive returns: the opposite of the leverage effect (Example 2)
 - ▶ QSD models, eq. (3): $f_{t+1} = \omega + \alpha \psi(y_t, X_t, f_t, \theta) + \beta f_t$, ψ the scaled score of an updating function ρ that need not equal $\log p$; theory: consistency, asymptotic normality, LR and LM tests with χ^2 limits (Section 3)
- **Estimation:** $\zeta = 0$ is kept inside the parameter space by eqs. (31)–(32): $\zeta \in (-1, 1/2)$ and $\Psi(x) = x(1 - e^{-cx})/(1 + e^{-cx})$ applied to the weight, $c = 1000$
 - ▶ PIT test [Diebold, Gunther & Tay \(1998\)](#): $u_t = F(\varepsilon_t)$ i.i.d. $U(0, 1)$ if the model is right; with the MLE and [Pierce \(1982\)](#), reject if $|\sum_t \hat{u}_t - T/2|/\sqrt{T(1/12 - B\mathcal{I}(\hat{\theta})^{-1}B^\top)} > 1.96$, $B = \lim_{T \rightarrow \infty} \mathbb{E}[\partial \sum_t (u_t - T/2)/\partial \theta^\top]$, $\mathcal{I}$ the Fisher information; it detects a shifted PIT mean, not every departure from uniformity
- **Data:** stocks with fewer than 4000 observations or with convergence problems are dropped: 400 stocks (Section 5)
 - ▶ Skewed variant, eq. (47): a skewed Student-t with asymmetry κ for ε_t and separate (ζ, τ) in the update; $\tau < 0.5$ gives a leverage effect whatever κ (Section 5.2)

Appendix: Case Study, an Example and the Limits Stated by the Authors

- Centene, 4330 days: $\hat{\xi} \approx 0.26$ (about 4 degrees of freedom) in all three models, but $\hat{\zeta} = 0.048$ (s.e. 0.027); LR 8.81 ($p = 0.003$) for $\xi = \zeta$ and 8.42 ($p = 0.004$) for $\zeta = 0$ (Table 5); after a crash day in March–April 2008, GARCH-t volatility stays well above Beta-t-GARCH and QSD lies between (Fig. 4)
- **Limits** stated by the authors
 - ▶ Beating GARCH-t and Beta-t-GARCH does not make the Student-t QSD model adequate: the three models fail the PIT test for 40.0–46.0% of the stocks (Table 4)
 - ▶ In the unmodified recursion (30), $\zeta < 0$ or $\zeta > 1/2$ does not define a proper volatility model, and $\zeta = 1/2$ with $\varpi = 0$ is degenerate (Section 4.1)
 - ▶ The $N(T/2, T/12)$ law of $\sum_t u_t$ holds only for known parameters; with estimated ones the PIT test needs the correction above (Section 5.1)
- **Open extensions**
 - ▶ Does the bounded QSD update beat GJR-GARCH-t in QLIKE out of sample, especially after March 2020? Does it tame the EUR/RON residual of 154? (Seminar 5, C3)

Appendix: The Discovery Loop: Reusable Prompts (1/2)

- Tools are interchangeable: an AI assistant (Claude, ChatGPT, Gemini, GitHub Copilot) and literature engines (Semantic Scholar, Elicit, Google Scholar)
- **Literature:** *"List out-of-sample comparisons of GARCH, HAR and machine-learning variance forecasts since 2015, with DOI, asset, loss function, test and horizon."*
 - ▶ Resolve every DOI on Crossref; check that the loss is robust to a noisy proxy [Patton \(2011\)](#)
- **Hypotheses:** *"For the claim that asymmetry improves one-day forecasts, list competing mechanisms (leverage, volatility feedback, a few crisis days) and what each predicts for d_t given the sign of r_{t-1} and by subperiod."*
- **Data and code:** *"Write Python with `arch` that re-estimates GARCH-t and GJR-t each January on an expanding window from data/market and stores one-day variance forecasts; no parameter may use data after the forecast origin."*
 - ▶ Replicate first: QLIKE 0.757 (GARCH-t) and 0.732 (GJR-t), DM $t = -1.22$

Appendix: The Discovery Loop: Reusable Prompts (2/2)

- **Robustness:** *“Write a pre-registered plan: splits by the sign of r_{t-1} and by subperiod, MSE besides QLIKE, the 22-day horizon, the HAC lag, and a Bonferroni or Model Confidence Set correction.”*
 - ▶ A random-split placebo is calibrated only if it reassigns blocks of days, preserving volatility clustering
- **Critique:** *“Act as a referee: could the proxy r_t^2 , one crisis year or the HAC lag create my result? Give the test that would expose each.”*
- **Report:** all splits and losses tried, adjusted p -values, code, seeds and package versions; a specification chosen after seeing results is exploratory and reported as such
- **Influential periods:** a crisis year can dominate $\bar{d}$; report subperiods and state in advance whether any year may be excluded

Appendix: The Team-Project Seed in Full

- Question: do machine-learning forecasts on daily data enter the Model Confidence Set together with GARCH, GJR and EGARCH for the S&P 500 and the BET, 2016–2026?
- Realised-variance (HAR) branch: the course folder `data/volare` holds 5-minute realised variances of 40 large American stocks, January 2015 – January 2026; there are no intraday data for the S&P 500 index or the BET, so this branch is run stock by stock
- Replicate first: the QLIKE table (GARCH- t 0.757, EWMA DM $t = 3.48$) and the mini-case splits
- Deliverables
 - ▶ A one-page pre-registration: models, losses, splits, error rate and what would falsify the claim
 - ▶ Code on `data/market`, the Model Confidence Set at 10%, a referee-style self-critique; `AI_USE.md` and `AI_ERRORS.md`
- Code of the mini-case: [Quantlets/Ch_05/ai_discovery_case.py](#)

Appendix: Further Resources

▣ Books

- ▶ [Franke, Härdle & Hafner \(2019\)](#), chapters on ARIMA and GARCH models for financial time series
- ▶ [Francq & Zakoïan \(2019\)](#): the mathematical theory of GARCH models

▣ Quantinar courses

- ▶  [Statistics of Financial Markets](#)
- ▶  [Applied Time Series Analysis with Python](#)

▣ Code

- ▶ All charts and tables: [Quantlets/Ch_05](#); Python package [Sheppard, arch](#)

References (1/6)

- ▣ Andersen, T.G., Bollerslev, T. (1998). Answering the skeptics: yes, standard volatility models do provide accurate forecasts. *International Economic Review*, 39(4), 885–905.
- ▣ Andrews, D.W.K. (2000). Inconsistency of the bootstrap when a parameter is on the boundary of the parameter space. *Econometrica*, 68(2), 399–405.
- ▣ Andrews, D.W.K. (2001). Testing when a parameter is on the boundary of the maintained hypothesis. *Econometrica*, 69(3), 683–734.
- ▣ Baillie, R.T., Bollerslev, T., Mikkelsen, H.O. (1996). Fractionally integrated generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 74(1), 3–30.
- ▣ Berkes, I., Horváth, L., Kokoszka, P. (2003). GARCH processes: structure and estimation. *Bernoulli*, 9(2), 201–227.
- ▣ Blasques, F., Francq, C., Laurent, S. (2023). Quasi score-driven models. *Journal of Econometrics*, 234(1), 251–275.
- ▣ Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327.
- ▣ Bollerslev, T. (1987). A conditionally heteroskedastic time series model for speculative prices and rates of return. *Review of Economics and Statistics*, 69(3), 542–547.
- ▣ Bollerslev, T., Wooldridge, J.M. (1992). Quasi-maximum likelihood estimation and inference in dynamic models with time-varying covariances. *Econometric Reviews*, 11(2), 143–172.

References (2/6)

- ▣ Christensen, K., Siggaard, M., Veliyev, B. (2023). A machine learning approach to volatility forecasting. *Journal of Financial Econometrics*, 21(5), 1680–1727.
- ▣ Clark, T.E., McCracken, M.W. (2001). Tests of equal forecast accuracy and encompassing for nested models. *Journal of Econometrics*, 105(1), 85–110.
- ▣ Creal, D., Koopman, S.J., Lucas, A. (2013). Generalized autoregressive score models with applications. *Journal of Applied Econometrics*, 28(5), 777–795.
- ▣ Diebold, F.X., Gunther, T.A., Tay, A.S. (1998). Evaluating density forecasts with applications to financial risk management. *International Economic Review*, 39(4), 863–883.
- ▣ Diebold, F.X., Mariano, R.S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.
- ▣ Engle, R.F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987–1007.
- ▣ Engle, R.F. (2004). Risk and volatility: econometric models and financial practice. *American Economic Review*, 94(3), 405–420.
- ▣ Engle, R.F., Ghysels, E., Sohn, B. (2013). Stock market volatility and macroeconomic fundamentals. *Review of Economics and Statistics*, 95(3), 776–797.
- ▣ Engle, R.F., Lilien, D.M., Robins, R.P. (1987). Estimating time varying risk premia in the term structure: the ARCH-M model. *Econometrica*, 55(2), 391–407.

References (3/6)

- ▣ Engle, R.F., Ng, V.K. (1993). Measuring and testing the impact of news on volatility. *Journal of Finance*, 48(5), 1749–1778.
- ▣ Fan, J., Qi, L., Xiu, D. (2014). Quasi-maximum likelihood estimation of GARCH models with heavy-tailed likelihoods. *Journal of Business & Economic Statistics*, 32(2), 178–191.
- ▣ Francq, C., Zakoïan, J.-M. (2004). Maximum likelihood estimation of pure GARCH and ARMA-GARCH processes. *Bernoulli*, 10(4), 605–637.
- ▣ Francq, C., Zakoïan, J.-M. (2019). *GARCH Models: Structure, Statistical Inference and Financial Applications*, 2nd ed. Wiley.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- ▣ Giacomini, R., White, H. (2006). Tests of conditional predictive ability. *Econometrica*, 74(6), 1545–1578.
- ▣ Glosten, L.R., Jagannathan, R., Runkle, D.E. (1993). On the relation between the expected value and the volatility of the nominal excess return on stocks. *Journal of Finance*, 48(5), 1779–1801.
- ▣ Hall, P., Yao, Q. (2003). Inference in ARCH and GARCH models with heavy-tailed errors. *Econometrica*, 71(1), 285–317.
- ▣ Hamilton, J.D., Susmel, R. (1994). Autoregressive conditional heteroskedasticity and changes in regime. *Journal of Econometrics*, 64(1–2), 307–333.

References (4/6)

- ▣ Hansen, B.E. (1994). Autoregressive conditional density estimation. *International Economic Review*, 35(3), 705–730.
- ▣ Hansen, P.R., Huang, Z., Shek, H.H. (2012). Realized GARCH: a joint model for returns and realized measures of volatility. *Journal of Applied Econometrics*, 27(6), 877–906.
- ▣ Hansen, P.R., Lunde, A. (2005). A forecast comparison of volatility models: does anything beat a GARCH(1,1)? *Journal of Applied Econometrics*, 20(7), 873–889.
- ▣ Hansen, P.R., Lunde, A., Nason, J.M. (2011). The model confidence set. *Econometrica*, 79(2), 453–497.
- ▣ J.P. Morgan/Reuters (1996). RiskMetrics — Technical Document, 4th ed. New York.
- ▣ Jensen, S.T., Rahbek, A. (2004). Asymptotic inference for nonstationary GARCH. *Econometric Theory*, 20(6), 1203–1226.
- ▣ Lamoureux, C.G., Lastrapes, W.D. (1990). Persistence in variance, structural change, and the GARCH model. *Journal of Business & Economic Statistics*, 8(2), 225–234.
- ▣ Lee, J.H.H., King, M.L. (1993). A locally most mean powerful based score test for ARCH and GARCH regression disturbances. *Journal of Business & Economic Statistics*, 11(1), 17–27.
- ▣ Li, W.K., Mak, T.K. (1994). On the squared residual autocorrelations in non-linear time series with conditional heteroskedasticity. *Journal of Time Series Analysis*, 15(6), 627–636.

References (5/6)

- ▣ Lumsdaine, R.L. (1996). Consistency and asymptotic normality of the quasi-maximum likelihood estimator in IGARCH(1,1) and covariance stationary GARCH(1,1) models. *Econometrica*, 64(3), 575–596.
- ▣ Mandelbrot, B. (1963). The variation of certain speculative prices. *Journal of Business*, 36(4), 394–419.
- ▣ McLeod, A.I., Li, W.K. (1983). Diagnostic checking ARMA time series models using squared-residual autocorrelations. *Journal of Time Series Analysis*, 4(4), 269–273.
- ▣ Mincer, J., Zarnowitz, V. (1969). The evaluation of economic forecasts. In J. Mincer (ed.), *Economic Forecasts and Expectations: Analysis of Forecasting Behavior and Performance*, NBER.
- ▣ Nelson, D.B. (1990). Stationarity and persistence in the GARCH(1,1) model. *Econometric Theory*, 6(3), 318–334.
- ▣ Nelson, D.B. (1991). Conditional heteroskedasticity in asset returns: a new approach. *Econometrica*, 59(2), 347–370.
- ▣ Patton, A.J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1), 246–256.
- ▣ Pele, D.T., Lazar, E., Dufour, A. (2017). Information entropy and measures of market risk. *Entropy*, 19(5), 226.
- ▣ Pele, D.T., Lazar, E., Mazurencu-Marinescu-Pele, M. (2019). Modeling expected shortfall using tail entropy. *Entropy*, 21(12), 1204.

References (6/6)

- ▣ Pele, D.T., Mazurencu-Marinescu-Pele, M. (2019). Using high-frequency entropy to forecast Bitcoin's daily Value at Risk. *Entropy*, 21(2), 102.
- ▣ Pierce, D.A. (1982). The asymptotic effect of substituting estimators for parameters in certain types of statistics. *The Annals of Statistics*, 10(2), 475–478.
- ▣ Self, S.G., Liang, K.-Y. (1987). Asymptotic properties of maximum likelihood estimators and likelihood ratio tests under nonstandard conditions. *Journal of the American Statistical Association*, 82(398), 605–610.
- ▣ Sheppard, K. [arch: ARCH models and other tools for financial econometrics in Python](#) (software documentation).
- ▣ West, K.D. (1996). Asymptotic inference about predictive ability. *Econometrica*, 64(5), 1067–1084.